# AC-SDKP Available for Research Use Only

## (No evidence-based human therapeutic dosing pattern)

AC-SDKP is an endogenous tetrapeptide derived from the N-terminus of thymosin beta 4. Repeated animal studies show antifibrotic and anti-inflammatory effects in cardiac, renal, hepatic, and pulmonary injury are likely. Angiogenic activity and reversible suppression of primitive hematopoietic progenitor proliferation are also established experimentally. Therapeutic efficacy from administered AC-SDKP in humans remains speculative.

AC-SDKP is N-acetyl-seryl-aspartyl-lysyl-proline, a naturally occurring four-amino-acid peptide also called goralatide. It corresponds to residues 1 through 4 of thymosin beta 4. Meprin-alpha and prolyl oligopeptidase participate in its release from thymosin beta 4, and the N-domain of angiotensin-converting enzyme rapidly degrades it. Its reported circulating half-life is very short at approximately 4.5 minutes.

💉 Short-term tolerability is likely for brief clinical exposure. Long-term safety is plausible, but sustained high exposure may affect erythropoiesis. Cancer and blood-cell caution: AC-SDKP promotes angiogenesis and can suppress primitive hematopoietic progenitor proliferation. These activities create plausible concerns in active malignancy and with sustained exposure in people vulnerable to anemia or marrow suppression.

⭐ ⭐ Likely effective for limiting pathological fibrosis and inflammatory remodeling based on consistent animal evidence across several organs. Human therapeutic efficacy remains speculative.

*Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.*

*This document was updated 12-Oct-26. The most recent version of this document is available at https://pdfs.researcher6076.com/pdfs/ac-sdkp.pdf*

## How AC-SDKP Fits Within Thymosin Beta 4 and TB-500 Biology

Full-length thymosin beta 4 contains multiple biologically active regions. AC-SDKP is the N-terminal 1-4 fragment associated primarily with antifibrotic, anti-inflammatory, angiogenic, and hematopoietic-regulatory activity. TB-500 is a different fragment, Ac-LKKTETQ, corresponding to residues 17 through 23 and centered on an actin-binding region. Evidence for TB-500 should not be attributed to AC-SDKP, and evidence for AC-SDKP should not be attributed to TB-500.

AC-SDKP is unusual among gray-market peptides because it is also an endogenous human metabolite and a recognized substrate of angiotensin-converting enzyme. ACE inhibitors raise plasma AC-SDKP concentrations several-fold in humans. This confirms human exposure to

elevated endogenous AC-SDKP but does not establish the efficacy of administering synthetic AC-SDKP.

## Dosing Patterns

No evidence-based human therapeutic dose or subcutaneous dosing interval has been established. The peptide degrades rapidly and most successful animal studies used continuous infusion by osmotic minipump rather than intermittent injections.

Translating those regimens into community injection schedules is speculative.

### Documented Clinical Exposure

**Historical goralatide trial:** A randomized oncology trial used 12.5 or 62.5 micrograms/kg/day on days 1 through 4 with carboplatin and fluorouracil. The study evaluated myeloprotection, not fibrosis treatment, and did not show hematologic benefit.

### Common Animal Research Exposure

**Continuous infusion:** Many rat studies used approximately 400 to 1,600 micrograms/kg/day delivered continuously by osmotic minipump.

## Combination Context

ACE inhibitors reduce AC-SDKP degradation and substantially raise circulating concentrations in humans. Concurrent synthetic AC-SDKP and ACE inhibition could plausibly increase or prolong exposure, while the magnitude and clinical consequences are unknown. In a 5xFAD mouse study, AC-SDKP was administered with TB-500 and the combination reduced neuroinflammatory and neuronal injury markers; the individual contribution of AC-SDKP was not isolated.

## Benefits

The most reproducible AC-SDKP findings involve limiting maladaptive inflammation, fibroblast activation, myofibroblast differentiation, and excess extracellular-matrix deposition. The evidence supports likely organ-protective effects in the animal injury models studied, while human therapeutic benefits remain speculative.

- **Cardiac antifibrotic effects:** AC-SDKP is likely to reduce cardiac inflammation, macrophage infiltration, and interstitial and perivascular fibrosis in hypertension, diabetes, myocardial infarction, and radiation injury models. Functional improvement varies by model.
- **Renal protection:** AC-SDKP is likely to reduce renal inflammation and fibrosis and may improve proteinuria or renal function in animal models of nephritis, reduced renal mass, ureteral obstruction, and diabetic nephropathy.


*This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076*

- **Hepatic antifibrotic effects:** AC-SDKP is likely to reduce hepatic inflammation, stellate-cell activation, collagen deposition, and TGF-beta/Smad signaling in animal models of chronic liver injury.
- **Pulmonary antifibrotic effects:** AC-SDKP is likely to reduce collagen deposition, myofibroblast differentiation, and epithelial-to-mesenchymal transition in animal models of silica-induced pulmonary fibrosis.
- **Inflammatory signaling regulation:** AC-SDKP is likely to reduce macrophage infiltration and inflammatory signaling in injured heart, kidney, liver, and lung tissue based on animal evidence.
- **Angiogenesis:** AC-SDKP is likely to stimulate endothelial migration, capillary-like structure formation, and new-vessel growth based on in-vitro and animal evidence.
- **Hematopoietic progenitor quiescence:** AC-SDKP reversibly suppresses proliferation of primitive hematopoietic progenitors in experimental systems. Clinical myeloprotection during chemotherapy remains speculative because a randomized trial found no meaningful hematologic benefit.
- **Neuroinflammatory support:** Reduced neuroinflammation, neurite atrophy, neuronal apoptosis, and memory impairment are plausible based on a combined AC-SDKP and TB-500 study in cell models and 5xFAD mice.

## Potential Applications

- **Cardiac remodeling:** AC-SDKP likely limits maladaptive inflammation and fibrosis after myocardial infarction, hypertension, diabetes, or cardiac irradiation in animal models.
- **Progressive kidney fibrosis:** AC-SDKP likely reduces renal inflammatory and fibrotic remodeling in several animal models. Human efficacy for chronic kidney disease remains speculative.
- **Chronic liver fibrosis:** AC-SDKP likely reduces fibrogenic signaling and collagen accumulation in experimental liver injury. Human efficacy remains speculative.
- **Fibrotic lung disease:** AC-SDKP likely reduces silica-induced pulmonary fibrosis in rats. Benefits for human silicosis or idiopathic pulmonary fibrosis remain speculative.
- **Ischemic tissue repair:** AC-SDKP plausibly supports microvascular repair through pro-angiogenic endothelial effects, while benefits depend on tissue and disease context.
- **Bone-marrow protection:** Protection of quiescent hematopoietic progenitors from selected cytotoxic injury is plausible from experimental evidence, while clinical myeloprotection remains speculative.


*This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076*

## Cautions

- **Active or previous malignancy:** Pro-angiogenic activity creates a plausible concern that AC-SDKP could support vascularization or progression of an existing malignancy. Its effects on cancer initiation and recurrence are speculative.
- **Anemia or marrow suppression:** AC-SDKP can inhibit primitive hematopoietic progenitor proliferation. Anemia and lymphopenia were more frequent in one short randomized oncology trial, and very high endogenous levels have been associated with erythropoietin resistance in renal failure.
- **ACE inhibitor use:** ACE inhibitors markedly increase endogenous AC-SDKP concentrations. Combined exposure could plausibly increase duration or magnitude of effect.
- **Impaired renal function:** Renal failure can substantially increase AC-SDKP accumulation, especially during ACE inhibition.
- **Pregnancy and breastfeeding:** Safety during pregnancy and breastfeeding is unknown. Avoid use as a precaution.

## Side Effects

- **Short clinical exposure:** A four-day randomized oncology trial described overall clinical and biological tolerability as excellent, but occurrences of anemia and lymphopenia increased.

## Potential for Diminishing Benefit

AC-SDKP is not established as a classical receptor agonist, and receptor desensitization has not been demonstrated. Its unusually short circulating half-life makes rapid enzymatic clearance a more immediate limitation than pharmacological tolerance.

## Biological Mechanism

AC-SDKP is released from the N-terminus of thymosin beta 4 through sequential proteolysis involving meprin-alpha and prolyl oligopeptidase. It is then hydrolyzed predominantly by the N-domain of ACE. ACE inhibition therefore raises endogenous AC-SDKP concentrations without requiring direct peptide administration.

Its antifibrotic activity is linked to reduced TGF-beta production or signaling, lower Smad2/3 and ERK1/2 activation, preservation or induction of inhibitory Smad7, reduced connective-tissue growth factor signaling, and suppression of fibroblast proliferation and myofibroblast differentiation. These pathways plausibly explain the repeated reductions in collagen and fibronectin deposition across organs.

Its anti-inflammatory effects include reduced macrophage infiltration, monocyte chemoattractant protein-1 activity, NF-kappa B signaling, and inflammatory cytokine expression


*This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076*

in experimental injury. AC-SDKP also acts directly on endothelial cells to promote migration, tube formation, and angiogenesis.

In hematopoiesis, AC-SDKP acts as a reversible negative regulator that keeps primitive progenitors out of the cell cycle. This can plausibly protect quiescent progenitors from cell-cycle-dependent injury, but excessive or sustained exposure could also suppress erythropoietic recovery.

## AC-SDKP and Cancer Risk

AC-SDKP stimulates endothelial migration and angiogenesis in experimental models. Because tumors can depend on neovascularization, support of existing tumor growth or progression is a plausible concern. AC-SDKP also alters hematopoietic progenitor cycling and has context-dependent effects in tumor and leukemia models. Its net effect on human cancer initiation, progression, treatment response, and recurrence remains speculative.

## AC-SDKP vs TB-500

TB-500 (Ac-LKKTETQ) and AC-SDKP (Ac-Ser-Asp-Lys-Pro) are distinct acetylated fragments of the 43-amino-acid peptide thymosin β4, corresponding to residues 17–23 and 1–4, respectively; neither is equivalent to full-length thymosin β4.

**Benefits and applications**: TB-500's experimental activity emphasizes cell migration, angiogenesis, and wound repair, supporting investigation in skin wounds and tendon injuries. AC-SDKP's strongest preclinical evidence concerns reducing inflammatory signaling, fibroblast activation, and excessive collagen deposition, supporting investigation in cardiac, renal, hepatic, and pulmonary fibrosis. Their activities overlap: both can promote angiogenesis, and TB-500's region also has experimental antifibrotic findings. The distinction is therefore an emphasis on wound repair versus limiting pathological fibrosis, rather than mutually exclusive effects. AC-SDKP additionally regulates primitive blood-cell progenitor proliferation, an activity relevant to its historical investigation as a marrow-protective agent.

**Evidence scope**: Direct TB-500 evidence consists principally of laboratory and animal studies; findings from full-length thymosin β4 cannot automatically be attributed to this fragment. AC-SDKP has a broader organ-fibrosis animal literature and limited historical human exposure as goralatide, but that exposure does not establish clinical antifibrotic efficacy. Human therapeutic benefits remain unproven for either fragment in these applications, and comparative clinical evidence does not establish that one is more effective or safer than the other.


*This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076*

## Conclusions

Short-term tolerability is likely for brief clinical exposure, while long-term safety is plausible. ACE inhibitor use, impaired renal function, anemia or marrow suppression, and malignancy materially change the biological context and warrant particular caution.

AC-SDKP is likely to reduce pathological inflammation and fibrosis in cardiac, renal, hepatic, and pulmonary animal models. Pro-angiogenic and hematopoietic-regulatory effects are also experimentally supported. Therapeutic human efficacy, intermittent subcutaneous dosing, and community cycling practices remain speculative.

## References

### Human Evidence

Cappelaere P, Hecquet B, Rolland F, et al. Randomized placebo trial of myeloprotection with goralatide in patients with squamous cell carcinoma of the upper respiratory and digestive tracts or esophagus treated with carboplatin and fluorouracil. Bull Cancer. 1995;82(9):732-737. PMID:8535033.

Azizi M, Ezan E, Reny JL, Wdzieczak-Bakala J, Gerineau V, Menard J. Renal and metabolic clearance of N-acetyl-seryl-aspartyl-lysyl-proline during angiotensin-converting enzyme inhibition in humans. Hypertension. 1999;33(3):879-886. doi:10.1161/01.HYP.33.3.879.

Azizi M, Ezan E, Nicolet L, Grognet JM, Menard J. High plasma level of N-acetyl-seryl-aspartyl-lysyl-proline: a new marker of chronic angiotensin-converting enzyme inhibition. Hypertension. 1997;30(5):1015-1019. doi:10.1161/01.HYP.30.5.1015.

Le Meur Y, Lorgeot V, Comte L, et al. Plasma levels and metabolism of AcSDKP in patients with chronic renal failure: relationship with erythropoietin requirements. Am J Kidney Dis. 2001;38(3):510-517. doi:10.1053/ajkd.2001.26839.

### Formation Metabolism and Mechanism

Kumar N, Yin C, Felcher P, et al. The anti-inflammatory peptide Ac-SDKP is released from thymosin-beta4 by renal meprin-alpha and prolyl oligopeptidase. Am J Physiol Renal Physiol. 2016;310(10):F1026-F1034. doi:10.1152/ajprenal.00562.2015.

Cavasin MA, Rhaleb NE, Yang XP, Carretero OA. Prolyl oligopeptidase is involved in release of the antifibrotic peptide Ac-SDKP. Hypertension. 2004;43(5):1140-1145. doi:10.1161/01.HYP.0000126172.01673.84.

Peng H, Carretero OA, Peterson EL, Rhaleb NE. Ac-SDKP inhibits transforming growth factor-beta1-induced differentiation of human cardiac fibroblasts into myofibroblasts. Am J Physiol Heart Circ Physiol. 2010;298(5):H1357-H1364. doi:10.1152/ajpheart.00464.2009.

### Cardiac and Vascular Evidence

Yang F, Yang XP, Liu YH, et al. Ac-SDKP reverses inflammation and fibrosis in rats with heart failure after myocardial infarction. Hypertension. 2004;43(2):229-236. doi:10.1161/01.HYP.0000107777.91185.89.

Sharma UC, Sonkawade SD, Spernyak JA, et al. N-acetyl-seryl-aspartyl-lysyl-proline treatment protects heart against excessive myocardial injury and heart failure in mice. J Am Heart Assoc. 2019;8(10):e012627. doi:10.1161/JAHA.119.012627.

Liu JM, Lawrence F, Kovacevic M, et al. The tetrapeptide AcSDKP, an inhibitor of primitive hematopoietic cell proliferation, induces angiogenesis in vitro and in vivo. Blood. 2003;101(8):3014-3020. doi:10.1182/blood-2002-07-2315.

Rhaleb NE, Peng H, Harding P, Tayeh M, LaPointe MC, Carretero OA. Effect of N-acetyl-seryl-aspartyl-lysyl-proline on DNA and collagen synthesis in rat cardiac fibroblasts. Hypertension. 2001;37(2 Pt 2):827-832.

### Renal Hepatic and Pulmonary Evidence

Omata M, Taniguchi H, Koya D, et al. N-acetyl-seryl-aspartyl-lysyl-proline ameliorates the progression of renal dysfunction and fibrosis in WKY rats with established anti-glomerular basement membrane nephritis. J Am Soc Nephrol. 2006;17(3):674-685. doi:10.1681/ASN.2005040385.

Wang M, Liu R, Jia X, Mu S, Xie R. N-acetyl-seryl-aspartyl-lysyl-proline attenuates renal inflammation and tubulointerstitial fibrosis in rats. Int J Mol Med. 2010;26(6):795-801. doi:10.3892/ijmm_00000527.

Chen YW, Liu BW, Zhang YJ, et al. Preservation of basal AcSDKP attenuates carbon tetrachloride-induced fibrosis in the rat liver. J Hepatol. 2010;53(3):528-536. doi:10.1016/j.jhep.2010.03.027.

Deng H, Xu H, Zhang X, et al. Protective effect of Ac-SDKP on alveolar epithelial cells through inhibition of EMT via TGF-beta1/ROCK1 pathway in silicosis in rat. Toxicol Appl Pharmacol. 2016;294:1-10. doi:10.1016/j.taap.2016.01.010.

Conte E, Fagone E, Fruciano M, et al. A new antifibrotic target of Ac-SDKP: inhibition of myofibroblast differentiation in rat lung with silicosis. PLoS One. 2012;7(7):e40301. doi:10.1371/journal.pone.0040301.

### Related Neuroinflammatory Evidence

Ou Z, et al. Thymosin beta4-derived peptides alleviate neuroinflammation and neurite atrophy in in-vitro models and 5xFAD mice: a potential therapeutic approach for memory improvement in Alzheimer's disease. Int Immunopharmacol. 2026:116097. doi:10.1016/j.intimp.2025.116097.